23.2: The Hardy-Weinberg equation can be used to test whether a population is evolving

The presence of genetic variation in a population does not guarantee that the population will evolve. For evolution to occur, one or more factors must be at work. In this section, we'll explore one way to test whether evolution is occurring in a population.

Gene Pools and Allele Frequencies

A **population** is a group of individuals of the same species that live in the same area and interbreed, producing fertile offspring. Different populations of a species may be isolated geographically from one another, exchanging genetic material only rarely. This is common for species that live on widely separated islands or in different lakes, but not all populations are isolated **(Figure 23.6)**. Still, members of a population typically breed with one another and thus are more closely related to each other than to members of other populations.

Figure 23.6

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One species, two populations.
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Porcupine herd of caribou ranges from northern Alaska and Canada bordering the Arctic Ocean; Forty mile herd range shares the southern reaches of the porcupine herd but travels further south.

These two caribou populations in the Yukon are not totally isolated; they sometimes share the same area (shaded purple). Still, members of each population are most likely to breed within their own population.

Porcupine herd of caribou range from northern Alaska and Canada bordering the Arctic Ocean; Forty mile herd range shares the southern reaches of the porcupine herd but travels further south. The porcupine herd is larger caribou with antlers, the Forty mile herd is smaller without antlers.

We can characterize a population's genetic makeup by describing its **gene pool**, which consists of all copies of every type of allele at every locus in all members of the population. If only one allele exists for a particular locus in a population, that allele is said to be *fixed* in the gene pool, and all individuals are homozygous for that allele. But if there are two or more alleles for a particular locus in a population, individuals may be either homozygous or heterozygous.

For example, imagine a population of 500 wildflower plants with two alleles, CR and CW, for a locus that codes for flower pigment. These alleles show incomplete dominance; thus, each genotype has a distinct phenotype. Plants homozygous for the CR allele (CRCR) produce red pigment and have red flowers; plants homozygous for the CW allele (CWCW) produce no red pigment and have white flowers; and heterozygotes (CRCW) produce some red pigment and have pink flowers.

A red flower is C superscript R, C superscript R; a white flower is C superscript W, C superscript W; a pink flower is C superscript R, C superscript W.

Each genotype and each allele has a frequency (proportion) in the population. To see how these frequencies are calculated, suppose our population has 320 plants with red flowers (genotype CRCR), 160 with pink flowers (genotype CRCW), and 20

with white flowers (genotype CWCW). Since 320 of the 500 plants have red flowers, the frequency of the CRCR genotype is 0.64 (320/500). Likewise, the frequency of the CRCW genotype is 0.32 (160/500) and the frequency of the CWCW genotype is 0.04 (20/500).

With respect to allele frequencies, wildflowers are diploid organisms, so each individual has two alleles for each of its genes. Hence, the 500 individuals in our population have a total of 1,000 copies of the gene for flower color. The CR allele accounts for 800 of these copies (320×2=640 for CRCR plants, plus 160×1=160 for CRCW plants). Thus, the frequency of the CR allele is 800/1,000=0.8 (80%).

When studying a locus with two alleles, the convention is to use p to represent the frequency of one allele and q to represent the frequency of the other allele. Thus, p, the frequency of the CR allele in the gene pool of this population, is p=0.8(80%). And because there are only two alleles for this gene, the frequency of the CW allele, represented by q, must be q=1−p=0.2 (20%). For loci that have more than two alleles, the sum of all allele frequencies must still equal 1100%.

The Hardy-Weinberg Equation

One way to assess whether natural selection or other factors are causing evolution at a particular locus is to determine what the genetic makeup of a population would be if it were *not* evolving at that locus. We can then compare that with the data we actually observed for the population. If there are no differences, this suggests that the population is not evolving. If there are differences, this suggests that the population is evolving—and then we can investigate why.

Hardy-Weinberg Equilibrium

In a population that is not evolving, allele and genotype frequencies will remain constant from generation to generation, provided that only Mendelian segregation and recombination of alleles are at work. Such a population is said to be in **Hardy-Weinberg equilibrium**, named for the British mathematician and German physician, respectively, who independently developed this idea in 1908.

To determine whether a population is in Hardy-Weinberg equilibrium, it is helpful to think about genetic crosses in a new way. Previously, we used Punnett squares to determine the genotypes of offspring in a genetic cross (see Figure 14.5). Here, instead of considering the possible allele combinations from one cross, we'll consider the combination of alleles in *all* of the crosses in a population. Imagine that all the alleles for a given locus from all the individuals in a population are placed in a large bin **(Figure 23.7)**. We can think of this bin as holding the population's gene pool for that locus. "Reproduction" occurs by selecting alleles at random from the bin; somewhat similar events occur in nature when fishes release sperm and eggs into the water or when plant sperm (in pollen) is blown about by the wind. By viewing reproduction as a process of randomly selecting and combining alleles from the bin (the gene pool), we are in effect assuming that mating occurs at random: that all male-female matings are equally likely.

Figure 23.7

Selecting alleles at random from a gene pool.

An illustration of the Hardy-Weinberg equilibrium of allele frequencies in a population.

Random selection of alleles from a gene pool following Hardy-Weinberg equilibrium shows that the gametes produced also follow the same ratios for allele frequency. The illustration is divided as follows:

- Frequencies of alleles:

The allele frequencies of the population are 0.8 (80 percent) and 0.2 (20 percent). p equals frequency of C superscript R allele (red) equals 0.8.

q equals frequency of C superscript W allele (white) equals 0.2.

- Alleles in the population:

If all the alleles could be placed in a large bin (representing the gene pool), 80 percent would be C superscript R and 20 percent would be C superscript W.

- Gametes produced:

Assuming mating is random, each time two gametes come together, there is an 80 percent chance the egg carries a C superscript R allele and a 20 percent chance it carries a C superscript W allele.

- Likewise, each sperm has an 80 percent chance of carrying a C superscript R allele and a 20 percent chance of carrying a C superscript W allele.

DRAW IT*Draw a bin with six white balls instead of four. For the frequency of*CR*to remain equal to*0.8*, how many red balls should the bin contain?*

*For suggested answer, see**Appendix A*.*

BioFlix® Animation

Allele Frequencies

image

image

Let's apply the bin analogy to the hypothetical wildflower population discussed earlier. In that population of 500 flowers, the frequency of the allele for red flowers (CR) is p=0.8, and the frequency of the allele for white flowers (CW) is q=0.2. This implies that a bin holding all 1,000 copies of the flower-color gene in the population would contain 800CR alleles and 200CW alleles. Assuming that gametes are formed by selecting alleles at random from the bin, the probability that an egg or sperm contains a CR or CW allele is equal to the frequency of these alleles in the bin. Thus, as shown in Figure 23.7, each egg (and each sperm) has an 80% chance of containing a CR allele and a 20% chance of containing a CW allele.

Using the rule of multiplication (see Figure 14.9), we can now calculate the frequencies of the three possible genotypes, assuming random unions of sperm and eggs. The probability that two CR alleles will come together is p×p=p2=0.8×0.8=0.64. Thus, about 64% of the plants in the next generation will have the genotype CRCR. The frequency of CWCW individuals is expected to be about q×q=q2=0.2×0.2=0.04, or 4%. CRCW heterozygotes can arise in two different ways. If the sperm provides the CR allele and the egg provides the CW allele, the resulting heterozygotes will be p×q=0.8×0.2=0.16, or 16% of the total. If the sperm provides the CW allele and the egg the CR allele, the heterozygous offspring will make up q×p=0.2×0.8=0.16, or 16%. The frequency of heterozygotes is thus the sum of these possibilities: pq+qp=2pq=0.16+0.16=0.32, or 32%.

As shown in **Figure 23.8**, the genotype frequencies in the next generation must add up to 1(100%). Thus, the equation for Hardy-Weinberg equilibrium states that if a population is not evolving at a locus with two alleles, the three genotypes will appear in the following proportions:

p2+2pq+q2=1ExpectedExpectedExpectedfrequencyfrequencyfrequencyof genotypeof genotypeof genotypeCRCRCRCWCWCW

Figure 23.8

Hardy-Weinberg equilibrium.

Illustration of Hardy-Weinberg equilibrium through generations of wildflowers.

In our wildflower population, the gene pool remains constant from one generation to the next. Mendelian processes alone do not alter frequencies of alleles or genotypes.

The gene pool of the initial (parental) generation consists of: 80% C superscript R (p = 0.8) crossed with 20% C superscript W (q = 0.2). A Punnett square is as follows:

- First row: C superscript R (p = 0.8) egg crossed with C superscript R (p= 0.8 sperm) = 0.64 (p squared) C superscript R, C superscript R. Phenotype: Red.

C superscript R (p = 0.8) egg crossed with C superscript W sperm (p = 0.2) = 0.16 (pq) C superscript R, C superscript W. Phenotype: pink.

- Second row: C superscript W (q = 0.2) egg crossed with C superscript R (p= 0.8 sperm) = 0.16 (q p) C superscript R, C superscript W. Phenotype = pink.

C superscript W (q = 0.2) egg crossed with C superscript W sperm (p = 0.2) = 0.04 (q squared) C superscript W, C superscript W. Phenotype = white.

- Genotypes in this offspring generation are: 64% C superscript R, C superscript R; 32% C superscript R, C superscript W and 4% C superscript W, C superscript W.
- Gametes in this offspring generation go to: 64% C superscript R (from C superscript R, C superscript R plants) + 16% C superscript R (from C superscript R, C superscript W plants) = 80% C superscript R = 0.8 = p.

4% C superscript W (from C superscript W, C superscript W plants) + 16% C superscript W (from C superscript R, C superscript W plants) = 20% C superscript W = 0.2 = q.

- With random mating, these gametes will result in the same mix of genotypes in the next generation: 64% C superscript R, C superscript R; 32% C superscript R, C superscript W and 4% C superscript R, C superscript W.

WHAT IF?If the frequency of theCR*allele wereo.6*, predict the frequencies of the*CRCR, CRCW, andCWCW*genotypes.*

*For suggested answer, see**Appendix A*.*

Figure 23.8 Walkthrough

Hardy-Weinberg equilibrium

Watch an explanation of this figure in the Figure 23.8 Walkthrough: Hardy-Weinberg equilibrium.

image

Note that for a locus with two alleles, only three genotypes are possible (in this case, CRCR, CRCW, and CWCW). As a result, the sum of the frequencies of the three genotypes must equal 1(100%) in *any* population regardless of whether the population is in Hardy-Weinberg equilibrium. The key point is that a population is in Hardy-Weinberg equilibrium only if the observed genotype frequency of one homozygote is p2, the observed frequency of the other homozygote is q2, and the observed frequency of heterozygotes is 2pq. Finally, as suggested by Figure 23.8, if a population such as our wildflowers is in Hardy-Weinberg equilibrium and its members continue to mate randomly generation after generation, allele and genotype frequencies will remain constant. The system operates somewhat like a deck of cards: No matter how many times the deck is reshuffled to deal out new hands, the deck itself remains the same. Aces do not grow more numerous than jacks. And the repeated shuffling of a population's gene pool over the generations cannot, in itself, change the frequency of one allele relative to another.

Conditions for Hardy-Weinberg Equilibrium

The Hardy-Weinberg approach describes a population that is not evolving. This can occur if a population meets all five of the conditions for Hardy-Weinberg equilibrium listed in **Table 23.1**. But in nature, the allele and genotype frequencies of a population often *do* change over time. Such changes can occur when at least one of the conditions for Hardy-Weinberg equilibrium is not met.

Table 23.1

Conditions for Hardy-Weinberg Equilibrium

Table 23.1, Conditions for Hardy-Weinberg Equilibrium .

The table is as follows:

Condition	Consequence if condition does not hold
1. No mutation	The gene pool is modified if mutations occur or if entire genes are deleted or duplicated.
2. Random mating	If individuals mate within a subset of the population, such as near neighbors or close relatives (inbreeding), random mixing of gametes does not occur and genotype frequencies change.
3. No natural selection	Allele frequencies change when individuals with different genotypes show consistent differences in their survival or reproductive success.
4. Extremely large population size	In small populations, allele frequencies fluctuate by change over time (genetic drift)
5. No gene flow	By moving alleles into or out of populations, gene flow can alter allele frequencies

Animation

Causes of Evolutionary Change

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Although departure from the conditions in Table 23.1 is common—resulting in evolutionary change—it is also common for natural populations to be in Hardy-Weinberg equilibrium for specific genes. One way this can happen is if selection alters allele frequencies at some loci but not others. In addition, some populations evolve so slowly that the changes in their allele and genotype frequencies are difficult to distinguish from those predicted for a non-evolving population.

CheckPoint

Using the Hardy-Weinberg Equation

Check your understanding before you move on, with the CheckPoint: Using the Hardy-Weinberg Equation.

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Applying the Hardy-Weinberg Equation

The Hardy-Weinberg equation is often used as an initial test of whether evolution is occurring in a population (Concept Check 23.2, question 3 is an example). The equation also has medical applications, such as estimating the percentage of a population carrying the allele for an inherited disease. For example, consider phenylketonuria (PKU), a metabolic disorder that results from homozygosity for a recessive allele. This disorder occurs in about one out of every 10,000 babies born in the United States. Left untreated, PKU results in mental disability and other problems. (As described in Concept 14.4, newborns are now routinely tested for PKU, and symptoms can be largely avoided with a diet very low in phenylalanine.)

To apply the Hardy-Weinberg equation, we must assume that no new PKU mutations are being introduced into the population (condition 1) and that people neither choose their mates on the basis of whether or not they carry this gene nor generally mate with close relatives (condition 2). We must also ignore any effects of differential survival and reproductive success among PKU genotypes (condition 3) and assume that there are no effects of genetic drift (condition 4) or of gene flow from other populations into the United States (condition 5). These assumptions are reasonable: The mutation rate for the PKU gene is low, inbreeding and other forms of nonrandom mating are not common in the United States, selection occurs only against the rare homozygotes (and then only if dietary restrictions are not followed), the U.S. population is very large, and populations outside the country have PKU allele frequencies similar to those seen in the United States.

If all these assumptions hold, then the frequency of individuals in the population born with PKU will correspond to q2 in the Hardy-Weinberg equation (q2=frequency of homozygotes). Because the allele is recessive, we must estimate the number of heterozygotes rather than counting them directly as we did with the pink flowers. Recall that there is one PKU occurrence per 10,000 births, which indicates that q2=0.0001. Thus, the frequency (q) of the recessive allele for PKU is

$q=0.0001=0.01$

and the frequency of the dominant allele is

$p=1-q=1-0.01=0.99$

The frequency of carriers, heterozygous people who do not have PKU but may pass the PKU allele to offspring, is

$2pq=2\times0.99\times0.01=0.0198$(approximately 2% of the U.S. population)

Remember, the assumption of Hardy-Weinberg equilibrium yields an approximation; the real number of carriers may differ. Still, our calculations suggest that harmful recessive alleles at this and other loci can be concealed in a population because they

are carried by healthy heterozygotes. The **Scientific Skills Exercise** provides another opportunity for you to apply the Hardy-Weinberg equation to allele data.

Scientific Skills Exercise

Using the Hardy-Weinberg Equation to Interpret Data and Make Predictions

AP Science Practices 2, 3, 4, 5, 6

Is Evolution Occurring in a Soybean Population? One way to test whether evolution is occurring in a population is to compare the observed genotype frequencies at a locus with those expected for a non-evolving population based on the Hardy-Weinberg equation. In this exercise, you will test whether a soybean population is evolving at a locus with two alleles, CG and CY, that affect chlorophyll production and hence leaf color.

Green soybeans.

How the Experiment Was Done Students planted soybean seeds and then counted the number of seedlings of each genotype at day 7 and again at day 21. Seedlings of each genotype could be distinguished visually because the CG and CY alleles show incomplete dominance: CGCG seedlings have green leaves, CGCY seedlings have green-yellow leaves, and CYCY seedlings have yellow leaves.

Data from the Experiment

A table of experiment results.

Time (days)	Number of Seedlings
Green (C superscript G C superscript G)	Green-yellow (C superscript G C superscript Y)
7	49
21	47

INTERPRET THE DATA

- Use the observed genotype frequencies from the day 7 data to calculate the frequencies of the CG allele (p) and the CY allele (q).
- Next, use the Hardy-Weinberg equation (p2+2pq+q2=1) to calculate the day 7 expected frequencies of genotypes CGCG, CGCY, and CYCY for a population in Hardy-Weinberg equilibrium.
-
  - Calculate the observed frequencies of genotypes CGCG, CGCY, and CYCY at day 7.
- Compare these frequencies to the expected frequencies calculated in question 2. Is the seedling population in Hardy-Weinberg equilibrium at day 7, or is evolution occurring? Explain your reasoning and identify which genotypes, if any, appear to be selected for or against.
-
  - Calculate the observed frequencies of genotypes CGCG, CGCY, and CYCY at day 21.

- Compare these frequencies to the expected frequencies calculated in question 2 and to the observed frequencies at day 7. Is the seedling population in Hardy-Weinberg equilibrium at day 21, or is evolution occurring? Explain your reasoning and identify which genotypes, if any, appear to be selected for or against.
- Homozygous CYCY individuals cannot produce chlorophyll. The ability to photosynthesize becomes more critical as seedlings age and begin to exhaust the supply of food that was stored in the seed from which they emerged. Develop a hypothesis that explains the data for days 7 and 21. Based on this hypothesis, predict how the frequencies of the CG and CY alleles will change beyond day 21.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.